Task 2: Predicting Variation in Emotional Valence and Arousal over Time from Ecological Essays

1. Introduction

Emotions are dynamic, they change over time; however, most of the current NLP research relies on static emotional data, such as social media posts. The main motivation of this project was to take the scope of NLP analysis into a dynamic domain, investigating the variation in emotions through ecological essays. At the end of the project, we created a model that is able to predict variation in emotions in real-world contexts and derive a long-term emotional analysis.

Several architectures, including GRU- and LSTM-based models, were explored during development; however, a RoBERTa–CNN architecture consistently outperformed recurrent approaches under the given data constraints.

Our dataframe was extremely limited (2700 samples), and we couldn't find any additional dataset that matched the dataset that the challengers gave. In addition to that, the performance of the embedding we went with, RoBERTa, was extremely dependent on the sample size, and it usually worked well with big sizes of data. Hence, we had to use almost all of our dataset to train the model, and we used the trial data that the challengers gave us to evaluate the model. We received 0.69 Macro F1-Score on Valence, and 0.72 Macro F1-Score on Arousal.

2. Related Work

The approach we followed required a system that involves RoBERTa tokenizer and Convolutional Neural Networks (CNNs). Hence, we've shaped our approach based on two articles that involved the use of models (including RoBERTa and CNN), and which hyperparameter combinations gave the best results. The design of our system is directly informed by prior work leveraging RoBERTa and hybrid transformer, CNN architectures for emotion prediction. Liu et al. introduce RoBERTa as a robustly optimized variant of BERT, demonstrating that improvements in pretraining strategies (such as dynamic masking, larger batch sizes, and longer training) lead to consistently stronger contextual representations across downstream NLP tasks, including sentiment and emotion analysis. Actually, the main motivation for using RoBERTa was the following: we had very limited data size, and training a Word2Vec embedding did not give the results we wanted, as it had too little data to learn the context from.

Building on this foundation, Mitsios et al. propose a RoBERTa-CNN architecture for emotion classification, showing that augmenting transformer outputs with convolutional

layers significantly improves performance. However, there was one problem before we procede with the approach that Mitsios et al proposed: they used three different datasets in their study: ISEAR, Wassa-21 and GoEmotions. Which means, they worked with large chunks of data, and they had more freedom to craft CNN architectures that has more layers, in other words, design a more complex system. The architecture they mentioned helped us to design our methodology a lot; however, we couldn't directly use the hyperparameters they've mentioned. Instead, we scaled them according to our dataset size and went with that.

We've researched recurrent architectures for continuous valence and arousal prediction, particularly GRU- and LSTM-based sequence models. For instance, Rasipuram et al. employ stacked GRU and BiLSTM layers to model temporal dependencies in affective signals and report competitive results on large-scale, multimodal datasets such as Aff-Wild2, which contain millions of frame-level annotations. In prior experiments, we used GRU-LSTM-based architectures, as the prior studies suggested. However, unlike the large-scale datasets used in prior studies, our dataset is comparatively small and text-only, which caused recurrent models to severely overfit, even though we used regularization and early stopping. Hence, we didn't go with this method.

Yu et al. introduce a weighted similarity graph approach that predicts valence and arousal ratings by propagating affective information across word-level semantic graphs constructed using Word2Vec embeddings. Graph-based similarity propagation methods relying on Word2Vec embeddings were also explored, but were ineffective due to insufficient data to learn meaningful lexical similarities.

Recent work on continuous valence–arousal estimation often employs the Concordance Correlation Coefficient (CCC) as a training objective, particularly in large-scale, multimodal, and temporally dense settings. For example, Lee et al. optimize a CCC-based loss to enforce temporal consistency and distributional alignment in frame-level VA prediction on the Aff-Wild2 dataset, where long sequences and large batch sizes are available. While CCC-based losses are commonly used for continuous valence–arousal prediction, we observed unstable optimization in our low-resource setting; therefore, we adopted a weighted MSE-based ordinal loss, which provided more reliable convergence.

3. Methodology

The dataset we're using is the one provided by the SemEval 2026 challengers. It had approximately 2700 samples, and the inputs were the texts that were written from the same user in different timestamps, the hour and the weekday (which are derived from the timestamp), and the collection phase. We've applied excessive research on the internet to

find a dataset that will comply with the challengers' one, however, none of them were in this format, and they were usually from visual or audial data, or texts from Twitter. Hence, unfortunately we're settled with the dataset that the challengers provided.

Our journey started with the GRU and LSTM classifiers, which was based on the approach that Rasipuram et al. had, because we thought treating the texts as sequence of tokens would help us detect variation in valence and arousal over time for the same user. The tokenization process was handled with a data cleaning pipeline, where we removed stopwords and repetitions, and converted emojis and emoticons to text. After the cleaning and tokenization process, we've trained Word2Vec embedding to capture contextual information about the tokens. We've tried different hyperparameter combinations, but as we were dealing with a small dataset, we usually kept skip-gram value at 1, and the context window at 5. We've also set the minimum word frequency to 1, because there were only 2700 sentences, and each unique word would count in capturing the context.

Small dataset size resulted in low performance on our Word2Vec embeddings, because it had very few tokens to learn the context similarities even though we've set minimum word frequency to 1. At this point, we've tried implementing similarity graph approach which was based on the prior study by Yu et al. More specifically, we've calculated the average valence and arousal levels that the tokens in each sentence by the user has. But once again, as the Word2Vec embedding couldn't derive the context similarities of words well, the results from the similarity graphs did not boost the performance of the model much.

At this point, we've searched for a methodology that has pretrained context similarities, and will not be affected by our small dataset size. The research we've made led us to integrate RoBERTa tokenizer into our system. The inspiration came from a previous study written by Ahire et al. as it explicitly mentions that RoBERTa is a robust verison of BERT and it enables stronger contextual representations. Although the study of Ahire et al. was extremely beneficial at understanding the importance of RoBERTa tokenizer, we based our implementation on another study, which is written by Mitsios et al.

Building on Mitsios et al., we implemented a hybrid RoBERTa–CNN model where RoBERTa provides contextual token embeddings and a lightweight CNN head captures local n-gram patterns that may signal affective shifts. In our setup, each input text was tokenized using the RoBERTa tokenizer and fed into a pretrained RoBERTa encoder. Then, we applied parallel 1D convolution filters with multiple kernel sizes over the sequence of hidden states, followed by max pooling to produce a compact text representation. In addition to this representation, we've also passed our derived numeric

features, which are hour, weekday, and collection phase, into the model and applied a small feedforward block to generate predictions.

The hyperparameter choices we adopted slightly differ from those reported by Mitsios et al., primarily due to the difference in dataset sizes. While Mitsios et al. employ a higher-capacity CNN head with at least 128 convolutional filters per kernel, resulting in a total CNN output dimensionality of 384 or more, we reduced the number of filters to 64 per kernel across kernel sizes 2, 3, and 4 (each represents an N-gram approach), yielding a 192-dimensional CNN representation. This design choice was made to limit model capacity and reduce the risk of overfitting given our dataset size of approximately 2,700 samples. We retained the RoBERTa-base architecture with 12 transformer layers and a hidden size of 768, as well as a maximum sequence length of 256, to preserve strong contextual representations. The pooled CNN outputs were combined with the RoBERTa layer and the numerical features, and passed through a feedforward network with hidden dimensions 256 and 128, combined with dropout regularization of 0.2 for valence, and 0.4 for arousal. For training, we followed a conservative setup with a batch size of 16, a learning rate of 2×10^{-5} for the RoBERTa encoder and 7.5×10^{-5} for the CNN and feedforward layers, along with early stopping. In addition to that, for the regression head, we've applied the learning rate scheduler to make sure that the validation loss keeps dropping. Finally, we employed 5-fold cross-validation and trained the model for 10 epochs in each fold. During each fold, the RoBERTa encoder was frozen for the first 4 epochs and unfrozen for the remaining 6 epochs to allow fine-tuning.

Another important nuance about our methodology is that we first framed the task as a regression problem rather than a direct classification problem. Both valence and arousal were predicted as continuous values using dedicated regression heads, and the resulting outputs were later mapped to discrete ordinal classes by rounding and clipping to the valid label ranges. This design choice allowed the model to explicitly capture the ordinal structure and distance relationships between emotion classes. During training, we employed a mean squared error (MSE) based ordinal loss, which was explicitly mentioned in the study by Mitsios et al, to ensure that prediction errors were penalized proportionally to their distance from the true label.

4. Results

Due to the limited size of the available dataset, all experiments were evaluated exclusively on the trial set that was provided by the challengers, which was used as a proxy test set to assess the performance and stability of the proposed models. In our research, we found out that RoBERTa is extremely dependent on the size of the data that we provide, so we've decided to use all 2700 of our samples in the training phase, and

test the performance of the model on the trial set, as the challengers didn't provide the test set yet. However, the trial set itself consisted of only 60 unseen samples, making the evaluation particularly volatile, where even a single misclassification could lead to noticeable fluctuations in performance metrics. To eliminate this instability and ensure low variation in results, we fixed the random seed across data loading, weight initialization, and training procedures.

Figure 1.1 and Figure 1.2 represent the classification results of valence and arousal respectively. An important result that can be derived is the following: the model performed well on classes that had more labels in the test set. For example in valence classification report, the classes "-2" and "2" has low support, meaning one misclassification on these classes change the Macro F1 score a lot. In other words, in the real performance indicators on a small test set, our model performed well. We were able to get 0.69 F1 score on valence, and 0.72 on arousal.

```
--- Valence Classification Report ---
              precision    recall  f1-score   support

          -2       1.00      0.50      0.67         6
          -1       0.72      1.00      0.84        13
           0       0.80      0.57      0.67        14
           1       0.70      0.90      0.79        21
           2       1.00      0.33      0.50         6

    accuracy                           0.75        60
   macro avg       0.85      0.66      0.69        60
weighted avg       0.79      0.75      0.73        60

Macro F1-Score:  0.6927
```

Figure 1.1 Classification Results of Valence

```
--- Arousal Classification Report ---
              precision    recall  f1-score   support

           0       0.67      0.50      0.57        16
           1       0.74      0.89      0.81        36
           2       1.00      0.62      0.77         8

    accuracy                           0.75        60
   macro avg       0.80      0.67      0.72        60
weighted avg       0.76      0.75      0.74        60

Macro F1-Score:  0.7169
```

Figure 1.2 Classification Results on Arousal

The distribution of predictions can be observed in more detailed from the confusion matrices in Figures 1.3 and 1.4, which represents valence and arousal respectively. An important nuance from the confusion matrix is that the model almost never misclassifies a label by more than two classes. Most of the errors we get on the test set are misclassified by one level. As we are working on ordinal labels, this means the models captures whether the valence value is good or not, but it's having difficulty in identifying the extent of the feeling. In other words, it misclassifies "good" with "very good".

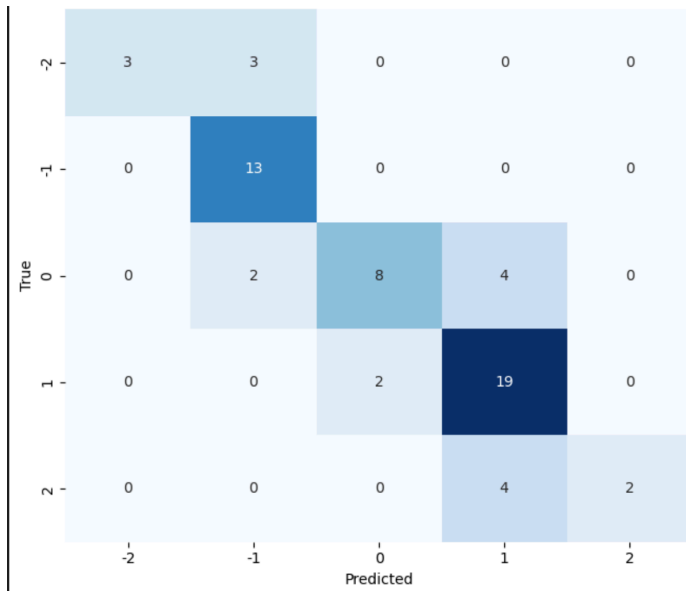

Figure 1.3 Valence Confusion Matrix

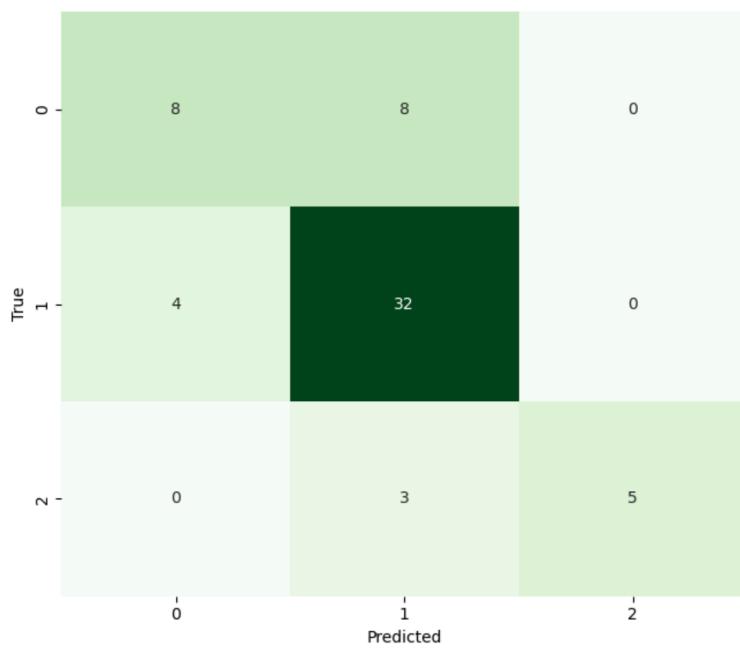

Figure 1.4 Arousal Confusion Matrix

In the appendix section, the precision and recall curves are provided in Figures 1.5 and 1.6, as well as the results and confusion matrices under our GRU-based approach. Figure 1.7 displays the scores that we were getting in the test set, and Figure 1.8 displays the confusion matrix for both valence and arousal. Unfortunately, we didn't keep track of the LSTM classifier results, but they were the same, as we were using Word2Vec embedding. Even though we were testing it with large test sets, the misclassifications were not low margin, and the model was not able to distinguish between classes well. Furthermore, that experiment taught us that we need to use all of our dataset in our context to get successful models.

When compared to the ordinal valence–arousal framework proposed in the referenced study, written by Mitsios et al., our findings exhibit consistent qualitative behavior despite the smaller evaluation set. Similar to the reported observations in the literature, most misclassifications in our model remain localized around neighboring ordinal classes, indicating that the model captures raw emotional structure rather than producing arbitrary errors.

5. Discussion

Overall, the results suggest that the proposed model exhibits consistent behavior on the trial set despite its limited size, achieving reasonable macro F1-scores while producing small margin ordinal errors. Although the small number of evaluation samples limits the conclusiveness of these findings, the observed error patterns indicate that the model captures valence–arousal distinctions mostly well.

One very important factor that may potentially caused harm on our model's performance is the dataset size. Even though we made deep research in many different sources to extend our data, we couldn't find any data source that is consistent with the dataset that the challengers provided us. Getting stuck to 2700 samples prohibited us from using more robust evaluations, and made us use our entire dataset.

Beyond dataset size, the choice of an ordinal valence–arousal formulation introduced both strengths and trade-offs. While this approach effectively constrained the model to make semantically plausible errors, it also reduced sensitivity to fine-grained emotional intensity, particularly for underrepresented classes. To be more specific, both of our valence and arousal datasets were dominated by the 0 class. Hence, the model had difficulties in capturing extreme classes like "-2" and "2" for valence, and "2" for arousal.

With additional time and resources, future work could focus on expanding the training data through carefully designed data augmentation or semi-supervised learning, as well as exploring alternative ordinal loss formulations that better handle class imbalance. Moreover, evaluating the system on a larger, officially released test set would allow for more reliable comparisons with existing methods and provide a clearer assessment of the model's generalization capabilities.

6. Conclusion

This work investigated the prediction of emotional valence and arousal variations from ecological essays under a low-resource setting. A RoBERTa–CNN model with an ordinal regression formulation achieved macro F1-scores of 0.69 for valence and 0.72 for arousal on the trial set, while mostly producing low-margin misclassifications. The results indicate that ordinal modeling helps capture raw affective structure even when fine-grained intensity distinctions are difficult due to limited data. Overall, the proposed approach provides a stable and interpretable baseline that can be further improved with larger datasets and more comprehensive evaluations.

7. Individual Contributions

Alp Eren Yelekçi: Handled all of the Exploratory Data Analysis (EDA) and did all the data-related research. Implemented the Word2Vec embedding and similarity graph.

Emre Çavuş: Handled all of the novel research, made explorations on adaptability and suitability of the approaches found in this task. Explicitly tried to find solutions to limited data problem, and made deep research about different word embeddings.

Abbas Kızıl: Initialized and constructed the models for GRU/LSTM classifiers as well as the CNN+RoBERTa architecture. Interacted with Emre Çavuş regularly on how to design the layer structures and arrange the embeddings.

Teoman Yamacı: Performed hyperparameter tuning on all of the approaches. Evaluated the results and came up with solution ideas. In addition to that, interacted with Emre Çavuş to base those ideas with the approaches in literature.

8. References

Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). *RoBERTa: A robustly optimized BERT pretraining approach*. arXiv. https://arxiv.org/abs/1907.11692

Mitsios, M., Vamvoukakis, G., Maniati, G., Ellinas, N., Dimitriou, G., Markopoulos, K., Kakoulidis, P., Vioni, A., Christidou, M., Oh, J., Jho, G., Hwang, I., Vardaxoglou, G., Chalamandaris, A., Tsiakoulis, P., & Raptis, S. (2024). *Improved text emotion prediction using combined valence and arousal ordinal classification*. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT 2024)* (pp. 808–813).

Tripathi, S., & Beigi, H. (2018). *Multi-modal emotion recognition on IEMOCAP dataset using deep learning*. arXiv. https://arxiv.org/abs/1804.05788

Yu, L.-C., Wang, J., Lai, K. R., & Zhang, X.-J. (2015). *Predicting valence-arousal ratings of words using a weighted graph method*. In *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics (ACL 2015)* (pp. 788–793).

Lee, J., Kim, S., Park, S., & Oh, A. (2019). *Prediction of continuous emotion using multi-task learning with concordance correlation coefficient loss*. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*.

9. Appendix

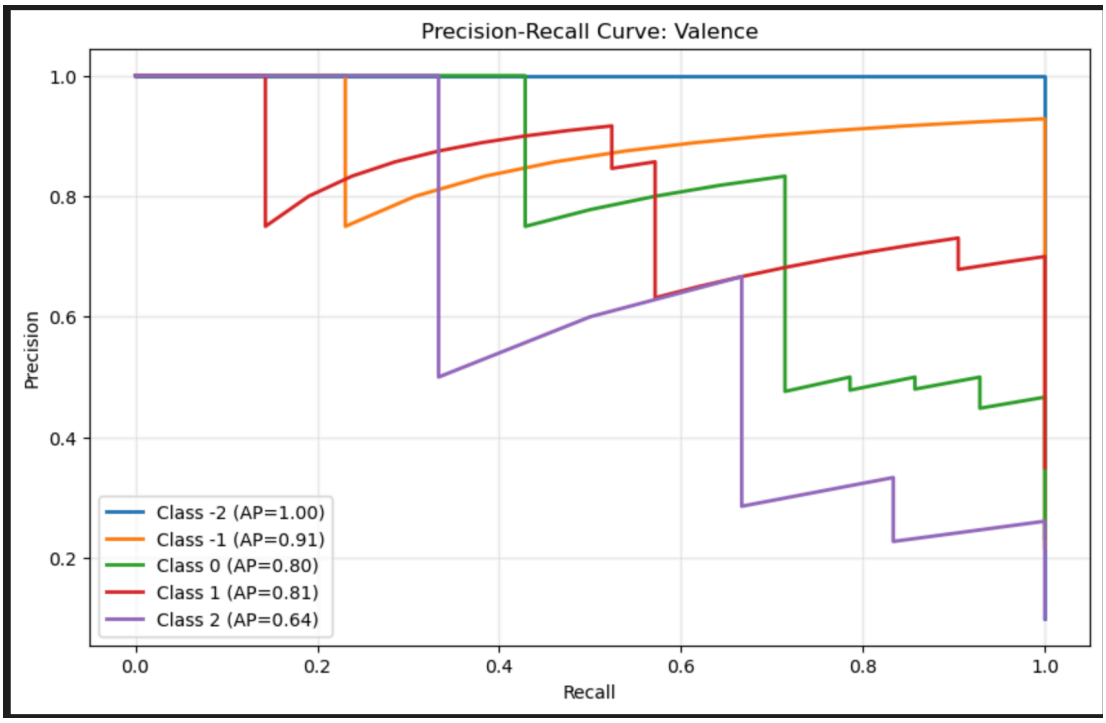

Figure 1.5 Precision-Recall Curve for Valence

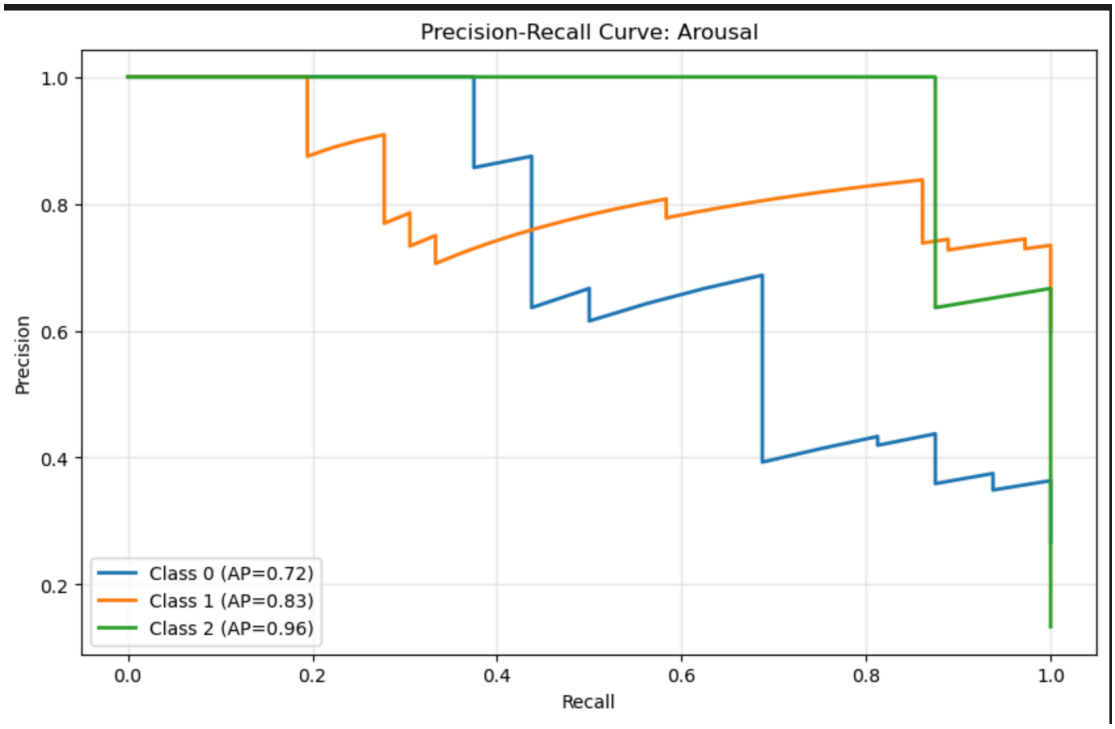

Figure 1.6 Precision-Recall Curve for Arousal

```
Test Valence:  acc = 0.409 | F1 = 0.390
Test Arousal:  acc = 0.517 | F1 = 0.468
```

Figure 1.7 Results on GRU Classifier

```
Valence confusion matrix (5 classes: 0..4):
[[16 15 10  2  2]
 [13 21 13  7  3]
 [18 17 80 19 13]
 [ 9  7 15 26 23]
 [ 4  1 10 25 41]]

Arousal confusion matrix (3 classes: 0..2):
[[122  33  14]
 [ 51  87  27]
 [ 29  15  32]]
```

Figure 1.8 Confusion Matrix of GRU classifier